SIX UNIT SQUARES CANNOT COVER A SQUARE OF SIDE GREATER THAN 2

NINGYU ZHOU

Abstract. Let $S(n)$ be the largest side length of a Euclidean square that can be covered by n closed unit squares, allowing arbitrary rigid motions and overlaps. Equivalently, Soifer's function $\Pi(2)$ is the least number of unit squares needed to cover *some* square of side $2+\varepsilon$ for *some* $\varepsilon > 0$. We present a computer-assisted *candidate* argument that $S(6) = 2$, or equivalently $\Pi(2) = 7$, addressing the conjecture stated by Dósa–Lángi–Tuza [4]. In particular, six unit squares cannot cover any square of side strictly greater than 2, while side 2 is attained by the obvious 2×2 tiling. The matching upper bound $\Pi(2) \le 7$ is the classical fact that seven unit squares cover some square of side > 2; see Soifer and the covering surveys cited below.

The argument is a finite case analysis on ownership of the centre and of the four outer midpoints, followed by stepwise normalisation / reclassification and the elimination of joining modes. Algebraic exclusions are discharged by exact rational identities, positive Bernstein certificates on compact boxes, interval certificates, and SMT UNSAT queries; selected interfaces are sketched in a Lean 4 skeleton. Machine certificates support the narrative; Lean Boolean flags are status metadata, not geometric theorems. Independent peer review has not occurred; see the repository status notes for open gaps.

Contents

Guide for referees 2
1. Introduction 2
2. Definitions and standing notation 3
3. Roles, normalisation, and parallel budgets 4
4. Top-level classification 5
5. Elimination of G5, G1, centre-G2, G3 and G4 5
6. Near-end residual G2 7
7. Far-end reduction to standard form 7

Date: October 21, 2026.

2020 *Mathematics Subject Classification.* Primary 52C15; Secondary 52C17, 68Q60, 68V15.

Key words and phrases. unit square covering, $S(n)$, $\Pi(n)$, computer-assisted proof, Bernstein certificates, SMT, Lean 4.

Computer-assisted candidate manuscript; generative tools used in production. The releaser does not claim mastery of every step; independent peer review is pending. See the repository `DISCLOSURE.md`.

8. Standard-form notation and four joining modes 8
9. Elimination of upper bridges and negative rotations 8
10. Right bridges are impossible 9
11. Double-direct positive rotations 10
12. Computational certificates 11
13. Related work 12
14. Concluding remarks 12
Acknowledgements 12
Appendix A. Closure patch M1: global normalisation honesty 12
Appendix B. Closure patch M2: frozen module R29N 13
Appendix C. Closure patch M3: explicit branch tables 13
Appendix D. Closure patch M4: E/H/S certificate tiers 14
References 14

Guide for referees

This short paper is a computer-assisted *candidate* argument, not a peer-reviewed resolution. Read in this order: (i) Definitions and Proposition 3.3 (normalisation honesty); (ii) Proposition 4.1 and Appendix C (branch tables); (iii) Propositions in Sections 5–11, each stating *hypothesis* $\Rightarrow$ *conclusion* and citing long-proof modules by ID; (iv) Appendix A–D (closure patches M1–M4); (v) repository `referee_pack/REFEREE_BRIEF.md` for timed reading paths. Boolean Lean flags are status metadata. Do *not* treat dual-solver UNSAT alone as a machine-independent proof.

1. Introduction

Covering a large square by equal smaller squares is a classical theme in discrete and metric geometry. Write

$$S(n) \;=\; \sup\bigl\{\, s>0 : \text{a closed square of side } s \text{ admits a covering by } n \text{ closed unit squares} \,\bigr\}, \tag{1}$$

where each unit square may be translated and rotated and overlaps are allowed. Soifer's function $\Pi(n)$ is the least number m of unit squares for which there *exists* some $\varepsilon>0$ such that a square of side $n+\varepsilon$ admits a covering by m unit squares (equivalently: there exists $\varepsilon_0>0$ such that every side in $(n, n+\varepsilon_0]$ admits such a covering); see [9, 8]. With this existential reading, and with the classical construction that seven unit squares cover some square of side strictly greater than 2 (hence $\Pi(2)\le 7$), one has

$$S(6)=2 \quad\Longleftrightarrow\quad \Pi(2)=7. \tag{2}$$

The lower bound $S(6)\ge 2$ is elementary: four axis-aligned unit squares tile $[0,2]^2$. The upper bound $S(6)\le 2$ is the content of the present paper; together with $\Pi(2)\le 7$ it would yield $\Pi(2)=7$.

Januszewski proved that covering a square of side $2+\varepsilon$ requires at least six unit squares, i.e. $\Pi(2) \geq 6$ [7]. Exact small values such as $S(5) = 2$ appear in the literature of Friedman–Paterson and Januszewski; see also Friedman's covering surveys [6, 5]. In January 2026, Dósa, Lángi and Tuza revisited congruent square coverings, determined several small $S(n)$ and boundary-covering variants, and stated as Conjecture 1.1 that $S(6) = 2$ (equivalently $\Pi(2) = 7$) [4].

Theorem 1.1 (Main claim)**.** $S(6) = 2$. *Equivalently (given* $\Pi(2) \leq 7$ *from known seven-square constructions),* $\Pi(2) = 7$*: six closed unit squares cannot cover any square of side strictly greater than* 2*, while a square of side* 2 *admits such a covering.*

The proof occupies Sections 2–11. Section 12 records the logical rôles of the computational certificates. Related work is collected in Section 13.

Remark 1.2 (Status)**.** Theorem 1.1 is offered as a *computer-assisted candidate*, supported by certificate inventories and a Lean 4 interface skeleton. Lean "no `sorry`" applies to that skeleton only; Boolean flags are *not* geometric theorems. Independent peer review has not occurred. Open gaps (in particular Lean classification fidelity) are listed in the repository file `referee_pack/KNOWN_GAPS.md`.

2. Definitions and standing notation

Definition 2.1. For each integer $n \geq 1$, $S(n)$ is defined by (1). Soifer's $\Pi(n)$ is the least integer m for which there *exists* some $\varepsilon > 0$ such that a square of side $n+\varepsilon$ admits a covering by m unit squares. Equivalently, there exists $\varepsilon_0 > 0$ such that every square of side in $(n, n+\varepsilon_0]$ admits such a covering. (The universal quantifier "for every $\varepsilon > 0$" is incorrect: for fixed m, large ε violate the area bound $(n+\varepsilon)^2 \leq m$.)

Throughout the upper-bound argument we work with the centred target

$$T \;=\; [-\mu, \mu]^2, \qquad \mu > 1, \tag{3}$$

and derive a contradiction from the assumption that six closed unit squares cover T. (The translate $[0, 2\mu]^2$ is equivalent for all length budgets.)

Write O for the centre of T, label the four corners $1, 2, 3, 4$ counter-clockwise from the lower left, and write BOTTOM, RIGHT, TOP, LEFT for the outer edges. An *outer mid* is the midpoint of an outer edge. The central horizontal and vertical target segments are denoted H and V. A *true gap* on an outer edge is a nonempty open subinterval of that edge not covered by the two adjacent corner tiles; endpoints of gaps are never free hypotheses and must be recovered from openness, convexity and closedness.

"Corner tiles abut" on an outer edge means that the two adjacent corner tiles together cover the entire edge (overlaps allowed). Otherwise the uncovered open interval is a true gap.

3. Roles, normalisation, and parallel budgets

Lemma 3.1 (Roles and midpoints)**.** *Four distinct tiles V_1, V_2, V_3, V_4 cover the four corners of T; a fifth tile C covers O; the sixth tile is denoted F. Each unit tile contains at most one outer mid. The role selection does not forbid F from containing a corner, nor a mid from lying in both a corner tile and a non-corner tile.*

Proof. Distinct target corners are at distance at least $2\mu > \sqrt{2}$, hence cannot lie in one unit tile. The distance from a corner to the centre is $\sqrt{2}\,\mu > \sqrt{2}$, so C is not a corner tile. Distinct outer mids are at least $\sqrt{2}\,\mu > \sqrt{2}$ apart. □

Lemma 3.2 (Dominated normalisation)**.** *Fix a target corner as origin with inward coordinates $X, Y \geq 0$. The unit tile containing that corner may be enlarged inside T to the form*

$$0 \leq cX + sY \leq 1, \qquad -\beta \leq -sX + cY \leq 1 - \beta, \quad c, s \geq 0,\ c^2 + s^2 = 1, \tag{4}$$

without moving any other tile and without deleting any previously covered point of T.

Proof. Among the four oriented unit normals choose one in the *closed* first quadrant (so $c, s \geq 0$; axis-aligned tiles have a normal on the boundary of the open first quadrant). Its original projection interval $[\alpha, \alpha + 1]$ contains 0, hence $\alpha \leq 0$. Replacing it by $[0, 1]$ does not remove any point of T with nonnegative projection. Keep the orthogonal unit interval unchanged. □

Proposition 3.3 (Dominated corner enlargement; stepwise reclassification)**.** *Starting from any covering of T, apply Lemma 3.2 to a corner tile V_i, leaving the other tiles fixed. The resulting covering still covers T and does not delete previously covered points of T. The same push applied a second time to that same tile is the identity on the already enlarged tile (idempotence of that single operator).*

This does not *imply that every later local normalisation or clamping step (for example, the $\beta \leq s/c$ clamp used for anchored corner tiles in the long verification manuscript) acts as the identity, nor that classification flags computed before an enlargement remain valid afterwards. After each enlargement that may create duplicate hosts or fill gaps, the covering must be* reclassified *on the new configuration; see the round 65 audit notes on β-clamping and stepwise reclassification in the ancillary package (`artifacts/still_S/S6_round65_full_audit.md`, §4).*

Lemma 3.4 (Two nonempty parallel chords)**.** *If a unit tile meets two parallel lines at distance $d > 1$, then $d \leq \sqrt{2}$ and the sum of the two chord lengths is at most*

$$B(d) = \begin{cases} d - \sqrt{d^2 - 1}, & 1 < d \leq 3/(2\sqrt{2}), \\ 2(\sqrt{2} - d), & 3/(2\sqrt{2}) \leq d \leq \sqrt{2}. \end{cases} \tag{5}$$

In particular $B(d) < 1$. Tangency counts as nonempty; an empty chord may not be substituted into the bound.

Proof sketch. With direction parameters $c \geq s > 0$ the transverse width is $w = c + s$. The two vertical lines lie at t and $t + d$ with $0 \leq t \leq w - d$. Since $d > 1 \geq c$, both chords fall in the triangular end-caps and the total length equals $(w-d)/(cs) = 2(w-d)/(w^2-1)$. Maximising over $d \leq w \leq \sqrt{2}$ yields the two branches of (5). The Euclidean chord-length identities matching the triangular-cap formulae are recorded in the Lean development of the parallel-budget geometry (Obligation 1). □

Further elementary budgets (grid contribution at most 2 per corner tile; $L + S + U < 2$ for a normalised corner tile; triangular area at most $1/2$) are collected as Lemmas R2–R3 in the long verification manuscript and used below without restating the proofs.

4. Top-level classification

If F also contains O, call the configuration class G5. Otherwise set

$$k \;=\; \#\{\text{outer mids belonging to } C \cup F\} \in \{0, 1, 2\}. \tag{6}$$

Then $k = 0$ is G1, $k = 1$ is G2, and $k = 2$ with adjacent (resp. opposite) mids is G4 (resp. G3). Each mid not yet supplied by $C \cup F$ chooses an adjacent corner host injectively. Reflection reduces host representatives.

Proposition 4.1 (Exhaustive branch table)**.** *Every covering of T falls into exactly one of the classes G5, G1, centre-owning G2, G4, G3, or residual G2 ($k = 1$ with the unique mid owned by F on* BOTTOM*). The residual G2 splits into a near-end host representative* $(\text{RIGHT}, \text{TOP}, \text{LEFT}) = (2, 3, 4)$ *and a far-end representative* $(2, 4, 1)$.

The global elimination order is: G5 for all three values of k; then G1, centre-owning G2, G4 and G3; then near-end residual G2; then reduction of the far-end representative to the final standard form; then elimination of all joining modes of that form.

Remark 4.2 (Branch tables)**.** Explicit tables (M3.0–M3.3) and the Boolean near-end checker appear in Appendix C and `artifacts/closure/`. The short-paper propositions below state hypothesis ⇒ conclusion and cite long-proof module IDs; they are not self-contained geometric proofs of every leaf.

5. Elimination of G5, G1, centre-G2, G3 and G4

Proposition 5.1 (G5 impossible)**.** Hypothesis. *F also contains O (class G5), with $k = \#\{\text{mid} \in C \cup F\} \in \{0, 1, 2\}$.* Conclusion. *No such covering exists; consequently in any covering of T exactly one tile contains O, and O is interior to that tile.*

Proof outline. Long-proof modules: R2 ($k = 2$), R3 ($k = 0$), R4–R8 ($k = 1$). For G5, modules R2–R3 eliminate $k = 2$ and $k = 0$. For $k = 1$, modules R4–R7 reduce to a triple-abutment configuration with an interior second centre

tile; module R8 then proves the two-corner inequality

$$L_1 + L_2 + \min(t_1, t_2) + \tfrac{4}{5}\max(t_1, t_2) \le \tfrac{14}{5} \tag{7}$$

by a Bernstein certificate on forty-five rational rectangles after mapping the full two-dimensional domain into the unit square, and compares it with the weighted three-ray bound on the interior tile to reach a contradiction in both orderings of (t_1, t_2). Thus G5 is empty. The remaining five tiles stay at positive distance from O, so a small neighbourhood of O is covered by C alone and O is interior to C. □

Proposition 5.2 (G1 impossible)**.** *If the centre tile is unique, then $C \cup F$ contains at least one outer mid.*

Proof outline. Module R9 eliminates triple abutment by a quarter-boundary budget and reduces to opposite or adjacent residual gaps. For opposite gaps, the corrigendum R10C first forces the horizontal chord of F to be strictly positive (via $g_1 + h_C > 1$ against $B(\mu) < 1$), then combines parallel-chord budgets of C and F with the corner contribution bound < 8. Adjacent gaps are eliminated in R10 §§3–6 by real endpoints and direction constraints. □

Proposition 5.3 (Centre cannot own the unique mid)**.** *If $k = 1$, the unique mid belonging to $C \cup F$ cannot belong to C.*

Proof outline. Axial propagation (R11) yields necessary outer contacts. Module R12 treats the two host representatives when only C owns the mid, covering all true-gap and triple-abutment patterns on LEFT, TOP and RIGHT. Shared ownership $C \cap F$ of the same mid is eliminated separately in R12 §10 by vertical redundancy and a true-gap triangle constraint. □

Proposition 5.4 (G4 impossible)**.** *No covering has C and F owning distinct adjacent outer mids.*

Proof outline. Normalise so that C owns the left mid and F owns the bottom mid. The three injective host representatives $(\mathrm{TOP}, \mathrm{RIGHT}) \in \{(4,3), (4,2), (3,2)\}$ are eliminated in modules R20–R23 by horizontal-sign analysis, forced suspension segments, and projection budgets on C. Shared left-mid ownership by the free corner tile is ruled out before the last representative. □

Proposition 5.5 (G3 impossible)**.** *No covering has C and F owning distinct opposite outer mids.*

Proof outline. Normalise so that C owns the left mid and F owns the right mid. Three host representatives (after reflection) are eliminated in R24–R25 by abutment/gap analysis on TOP and BOTTOM, empty/nonempty vertical chords of F, and a local exit lemma for nonempty cases. □

Corollary 5.6 (Universal centre forbids outer mids)**.** *In any covering of T the unique centre tile contains no outer mid, and the remaining non-corner tile contains some outer mid.*

Proof. Combine Propositions 5.1–5.5. □

6. Near-end residual G2

After Corollary 5.6 the only surviving class is residual G2 with the unique mid owned by F on BOTTOM.

Proposition 6.1 (Near-end impossible)**.** Hypothesis. *Residual G2 with F owning the bottom mid, C owning no outer mid, and host injection* $(\mathrm{RIGHT}, \mathrm{TOP}, \mathrm{LEFT}) = (2, 3, 4)$ *(free corner $W = V_1$).* Conclusion. *No such covering of T exists.*

Proof outline. See long-proof modules R13–R19 and R26–R28, summarised by the Boolean branch table in Appendix C (Table M3.1). Module R18 shows that either both RIGHT and TOP abut, or exactly one of them carries a true gap owned by C. In the double-abutment case: R13 eliminates nonempty horizontal chords of F; R14–R15 force left abutment when that chord is empty and produce an interior point Q_0 of C; R15S/R16P supply the strict angular separation and height hypotheses; R16 then covers the necessary continuous domain by 1173 rational boxes with interval certificates excluding every leaf. Remaining left-gap / nonempty-F patterns are eliminated in R17–R18. Top-gap patterns fall to R19 and R26; right-gap patterns fall to R27–R28 (both rotation signs). A normalisation that fills a gap returns to an already eliminated abutment leaf (Appendix A). □

7. Far-end reduction to standard form

Henceforth fix the far-end representative: V_1 owns the left mid, V_2 the right mid, V_4 the top mid, and the free corner tile is $W = V_3$. The centre tile C owns no outer mid; F owns the bottom mid and not the centre.

Proposition 7.1 (Far-end standard form)**.** Hypothesis. *Residual G2 with F owning the bottom mid, C owning no outer mid, and far-end hosts $V_1 \supset$* LEFT mid*, $V_2 \supset$* RIGHT mid*, $V_4 \supset$* TOP mid*, free corner $W = V_3$.* Conclusion. *The covering satisfies:* LEFT*,* RIGHT *and* TOP *abut via corner tiles; $F \subset \{y < 0\}$; C avoids* BOTTOM*; and one copy of F contains both closed endpoints of the bottom gap together with the lower axial tip of C (the* final standard form*).*

Proof outline. See Table M3.2 in Appendix C and frozen module R29N in Appendix B.

Step 1. Modules R27 and R29 eliminate standardised top true gaps for both rotation signs of C. Thus TOP abuts. Any lateral true gap lies too far from the bottom mid for F to reach, so lateral gaps are owned entirely by C. Since C cannot meet both LEFT and RIGHT at distance $2\mu > \sqrt{2}$, the only remaining cases are left gap, right gap, or triple abutment (Proposition R29N.1).

Step 2. If $F \cap H \neq \varnothing$, Proposition R29N.2 forces C to avoid BOTTOM; R31 eliminates triple abutment and R32–R36 eliminate the lateral-gap cases. Hence $F \cap H = \varnothing$. If some point of F had $y \geq 0$, convexity with $(0, -\mu) \in F$ would produce a point of $F \cap H$ inside T, a contradiction. Thus $F \subset \{y < 0\}$.

Step 3. Corollary 5.6 applied to an inscribed target yields axial-root caps $\leq 2-\mu$ (module R37) without circularity. Modules R37–R39 eliminate all right and left true-gap patterns (both rotation signs, direct and bridged contacts).

Step 4. In the residual triple-abutment case, R39 gives the upper axial root $u > 38/79$, whence C avoids BOTTOM. The three anchored corner tiles can contribute only their matching central half-axes, so the entire bottom gap and the lower tip of C are forced into the same tile F. □

8. Standard-form notation and four joining modes

Let $a, x, z \in (0,1)$ be the half-angle parameters of V_1, V_2, V_4, and let $h \in [0, \sqrt{2}-1]$ be the half-angle of the smaller axial angle of C. Set

$$b(t) = \frac{1+t^2}{1-t^2}, \quad k(t) = \frac{2t}{1-t^2}, \quad g(t) = \frac{1-t}{1+t}, \quad q(t) = \frac{t}{2-t}, \tag{8}$$

and write $\delta(t) = b(t) - 1$, $\Delta_z = \delta(z)$. The four actual axial tips of C are $L = (-l, 0)$, $R = (r, 0)$, $T_C = (0, u)$ and $B_C = (0, -d)$. Writing S_1, S_2 for the actual short legs of the bottom corner tiles and $\Sigma = S_1 + S_2$, Proposition 7.1 yields

$$\Sigma > 2 - \sqrt{2} > \tfrac{7}{12}, \qquad \Sigma > (1-d)^2, \qquad \Sigma > 2 - \frac{1}{1-d}. \tag{9}$$

Left abutment supplies the constraints

$$0 < z < a < 1, \quad a+2z < 1, \quad z < \tfrac{4}{15}, \quad 0 \leq S_1 < \mathcal{F}(a,z) := b(a) - k(a)\big(2 - g(z)\big). \tag{10}$$

Lemma 8.1 (Normalised seam point in C)**.** *There is a point*

$$P_1 = (-1 + X, e) \in C, \qquad X = \frac{b(a) + S_4 - 2}{k(a) - k(z)}, \quad e = 1 - S_4 - k(z)X, \tag{11}$$

obtained from the real-scale seam $P_\mu \in C$ by a convexity argument inside the same tile C (modules R40 and R42). The construction does not rescale the covering to $\mu = 1$ and does not move C.

Let Q_2 and Q_U be the actual inward tips of V_2 and V_4 on the positive axes. Whether C contains these tips yields four exhaustive joining modes: double-direct, upper bridge by W, right bridge by W, and double bridge. If C misses a tip, the corresponding open axial gap can be filled only by the same free corner tile W, which is then forced by closedness to contain both the tip of C and the corner tip.

9. Elimination of upper bridges and negative rotations

Proposition 9.1. *In the standard form of Proposition 7.1, the free corner tile W cannot contain Q_U.*

Proof outline. Modules R44–R45 show that an upper bridge forces $z \le \tau < 2/5$ for the parameter τ of W and empties $W \cap H$, whence C contains Q_2 directly and double bridging is impossible. Modules R46 and R48 produce a real exit point $T_R \in C \cap W \cap V_2$. Small angles $h \le 1/24$ fall to R52. Positive rotations are eliminated in R53–R55 via the seam-derived bound $S_1 < q(Q)$ and the two normals of C at T_R. Negative rotations are reduced in R56–R58 to $0 < \tau < 1/6$ and then excluded in R59 by the area-forced negativity

$$G = H_F - q(x+\delta) - g(x) + k(x)(B_t - 1) < 0, \quad H_F = 2 - \frac{1}{1-\delta-\tau}, \tag{12}$$

using endpoint polynomial certificates on small/medium x and a left-root projection contradiction on large x. □

Consequently only double-direct and right-bridge modes remain, both with $C \ni Q_U$ and hence $u > 1 - z$. In the right-bridge case the horizontal chord length of W is strictly less than Δ_z, which after establishing that the relevant parameters are < 1 yields

$$S_2 < q(r + \Delta_z); \tag{13}$$

double-direct admits the stronger $S_2 < q(r)$.

Proposition 9.2. *All negative rotations in the two residual joining modes are impossible. Every surviving positive rotation satisfies* $1/10 < z < 1/5$.

Proof outline. Small-angle double-direct and right-bridge cases are eliminated in R43 and R49–R51 respectively, both forcing $h > 1/8$. Module R60 partitions the negative-rotation domain into four regions and contradicts (9) in each, paying the loss (13). Modules R61–R62 eliminate $z \ge 1/5$ and $z \le 1/10$ for positive rotations by asymmetric triangles on the same F and concavity of the budget form, leaving the open strip $1/10 < z < 1/5$. □

10. Right bridges are impossible

Proposition 10.1. *In the residual domain of Proposition 9.2, W does not contain the right axial tip R of C. Consequently the entire right-bridge class is empty.*

Proof outline. Module R63 first obtains $r < 17/20$ from axial chords and the R/T_C projection. Assuming $W \ni R$, the tile meets both TOP and H, forcing a top leg $D > 1 - \Delta_z > 11/12$ and a half-angle

$$t < \tfrac{1}{8}, \qquad t < \tfrac{9}{7}\Delta_z. \tag{14}$$

Comparing real projections in both orientations of W yields an abscissa ξ of the contact on H satisfying

$$\xi < \Delta_z + t \quad \text{or} \quad \xi > 1 - t > \tfrac{7}{8}. \tag{15}$$

The second alternative contradicts $r < 17/20$ if the contact is R. The first, combined with (13)–(14) and left long-leg loss, produces

$$\Sigma < \Phi\Big(\frac{2z}{1+z}\Big) + q\Big(\frac{23}{7}\Delta_z\Big), \quad \Phi(w) = (1+w)\big(1-\sqrt{w(2+w)}\big), \tag{16}$$

which is decreasing on $1/10 \le z \le 1/5$ and strictly less than $7/12$ at the left endpoint, contradicting (9). □

11. Double-direct positive rotations

Proposition 11.1. *No positive-rotation double-direct covering survives in the residual strip of Proposition 9.2.*

Proof outline. From (10) and $1/10 < z < 1/5$ one obtains $2/7 < a < 3/5$. Double-direct gives $S_2 < q(r)$. At the formal endpoint $S_4 = g(z)$ define

$$X_G = \frac{(1-z)\big[a^2(1+2z) - z\big]}{(a-z)(1+az)}, \qquad e_G = \frac{2az(1-a-z)}{(a-z)(1+az)}. \tag{17}$$

After verifying $a < g(h)$ one has the projection bound

$$r < R_P(h) := \delta(h) + X_G + k(h)e_G, \tag{18}$$

while the pair (T_C, R) yields $r < R_T(h) := h + z/k(h)$ for $h > 0$.

On $0 \le h \le 1/4$ one has $a < 3/5 \le g(h)$; monotonicity of (18) reduces the right-root upper bound to $h = 1/4$, after which $\mathcal{F} + q\big(R_P(1/4)\big) < 7/12$ on the full (a, z)-rectangle. On $1/4 \le h \le 1/3$, concavity of the real P_1/T_C relation forces $a > 2/5$, and four endpoint values control $\Sigma < 7/12$. On $1/3 \le h \le \sqrt{2} - 1$, a rational outer cap $29/70$ splits according as $a \ge g(h)$ or not; both subcases again force $\Sigma < 7/12$.

The fourteen deduplicated rational targets are certified by twenty-eight numerator/denominator polynomials whose one hundred fourteen strictly positive rational Bernstein coefficients, together with reconstruction identities, cover the whole continuum $[0, \sqrt{2} - 1]$ (module R64). Every angular segment contradicts (9). □

Proof of Theorem 1.1. Assume six unit squares cover T with $\mu > 1$. Proposition 4.1 places the covering in one of the top-level classes. Propositions 5.1–5.5 eliminate G5, G1, centre-owning G2, G3 and G4. Proposition 6.1 eliminates the near-end residual G2. Proposition 7.1 reduces the far-end residual G2 to the final standard form. Propositions 9.1–11.1 eliminate every joining mode, rotation sign and residual parameter domain of that form. Hence (subject to the completeness of the cited modules and certificates) $\mu > 1$ is impossible under the candidate narrative. Combined with the elementary covering of side 2 and the classical $\Pi(2) \le 7$, this would yield $S(6) = 2$ and therefore $\Pi(2) = 7$ by (2). The public argument remains a candidate pending independent verification. □

12. Computational certificates

The geometric narrative of Sections 3–11 is supported by three layers of evidence.

Exact rational certificates. On a polynomial over $[0,1]^m$, the tensor Bernstein basis is nonnegative and sums to one. Nonnegative (resp. strictly positive) coefficients prove nonnegativity (resp. strict positivity). Parameter domains are mapped into rectangles by explicit rational substitutions after denominators are shown positive. Verifiers regenerate targets, rebuild basis-change identities and rational coefficients, rather than trusting a generator's claimed minimal coefficient. Module R8 additionally checks a complete prefix-tree cover; module R64 checks every polynomial on its fixed rectangles.

Continuous interval certificates. The R16 root box outwardly contains the necessary domain. Each internal node splits into children whose union recovers the parent; each leaf is excluded by exact fractional envelopes. Denominator positivity, monotonic substitution direction, and all leaves together with their boundaries are checked.

Real-algebraic UNSAT queries. Selected propositions are accompanied by QF_LRA or QF_NRA queries. Only UNSAT is accepted. Queries are classified as (E) exact certificates, (H) hand proofs with solver cross-check, or (S) dual-solver UNSAT (z3 and cvc5) not yet reduced to (E). Dual-wrapper spot checks on the M8–M11 inventory report pass on the expanded faithfulness sample shipped with the ancillary package. Finite floating-point sampling is never used as an exclusion certificate.

Lean 4 fragments. Selected interfaces—parallel-budget geometry and Euclidean chord lengths; partial ownership / role bridges; M7.1 branch-logic scaffolding; far-end PDF faithfulness sketches; and algebraic pieces used in M8–M11 together with near-end leaf wirings—are present in the Lean 4 package `S6Core` (mathlib). The shipped skeleton builds with no `sorry` and no extra axioms *as a skeleton*. Boolean "closed" flags (e.g. `wholeTheorem_S6_eq_2_closed`) are human-maintained *status metadata*, not theorems about six unit squares covering a target. Several leaf conclusions are still propositional placeholders of the form `living` $\to$ `True`; ownership-bridge classifiers are not yet proved faithful to the manuscript's k / $F \ni O$ / sixth-tile role selection. The long verification manuscript remains the geometric reference for metric inequalities discharged primarily by certificates.

Reproduction. The public repository ships Lean sources under `lean/`, certificate registries and spot-check checkers under `artifacts/`, and the long verification manuscript under `artifacts/long_proof/` (`S6_full_proof.md`/`.pdf`). Companion certificate JSON files needed by the shipped R42/R54–55/R64 checkers sit next to those scripts. A compressed copy of the historical `ex/all/` verification tree is shipped as `artifacts/ex_all.tar.gz` with SHA-256 sums in `artifacts/SHA256SUMS.txt`; an unpacked copy sits at `artifacts/ex/all/`.

Closure patches M1–M4 live in `artifacts/closure/` and in Appendices A–D. See `referee_pack/HOW_TO_REPRODUCE.md`. A minimal path from a clean clone is

```
cd artifacts/checkers && python3 check_round64.py && python3 check_round63.py
cd ../closure && python3 m3_branch_checker.py
# optional: cd ../../lean/S6Core && lake build
```

13. Related work

Soifer's JCTA note and Cover-Up Squared writings introduced $\Pi(n)$ [9, 8]. Januszewski's Monthly note established $\Pi(2) \geq 6$ [7]. Friedman–Paterson and Friedman's surveys record constructions and small-case discussions [6, 5]. Dósa–Lángi–Tuza proved several exact small values and boundary analogues and isolated the conjecture $S(6) = 2$ [4]; Theorem 1.1 is a candidate argument addressing that conjecture. Asymptotic packing-waste problems of Chung–Graham [2, 3] concern a different regime.

On computer-assisted proofs in geometry, we follow the AMS guidelines on independent checkability and durable availability of code and data [1].

14. Concluding remarks

If confirmed, Theorem 1.1 would determine the first previously open value of $S(n)$ beyond the classical small cases. The same case-analysis and certificate pipeline suggests approaches to neighbouring values (e.g. refinements of $\Pi(3)$ and boundary-covering variants $S_{\mathrm{bd}}(n)$ from [4]), though those questions lie outside the present paper.

Acknowledgements

Certificate archives and Lean scaffolding accompany the argument in the public repository. Production used computer assistance (including generative tools); the releaser does not claim mastery of every step. See `DISCLOSURE.md`.

Appendix A. Closure patch M1: global normalisation honesty

The maturity revision weakened Proposition 3.3: a single dominated enlargement is idempotent, but later β-clamps need not be the identity, and classification must be recomputed after enlargements that fill gaps. The following propositions, adapted from the Chinese closure note `artifacts/closure/M1_proposition_zh.m` record the stronger *intended* global protocol used by the long proof. They are packaging and logical-cycle closures, not new geometric exclusions.

Proposition A.1 (One-shot global corner normalisation)**.** *Apply Lemma 3.2 once to each corner tile $V_1, \ldots, V_4$, leaving C and F fixed, and (when $c, s > 0$) clamp the orthogonal parameter β into the interval determined by s/c and $1 - c/s$ as in the long manuscript. Then: coverage of T is nondecreasing; roles of C and F are unchanged up to non-increasing k; open-first-quadrant*

normals are unique so the procedure is canonical; and no corner tile acquires O or two distinct outer mids.

Proposition A.2 (Classify once after global normalisation)**.** *On the output of Proposition A.1, perform the G1–G5 / host / near–far / abutment–gap / $F \cap H$ classification of Section 4 exactly once. Subsequent local "normalisation" steps of types (Lα) and (Lβ) on already-normalised corner tiles are the identity, so there is no "normalise → reclassify → normalise" loop inside elimination modules.*

Remark A.3. Proposition A.2 assumes no module requires a strictly *non*-normalised corner tile. The module-by-module check is recorded in `artifacts/closure/REVIEW_addend` If a future module needs a non-normalised intersection, that module must be rewritten or the global protocol delayed with a well-founded order. Until independently verified, treat Propositions A.1–A.2 as candidate closure text accompanying the long proof.

Appendix B. Closure patch M2: frozen module R29N

References in the short paper and long proof to "round-29 continuation notes §4" are hereby frozen as module **R29N**. Withdrawn note §5 (swapped parallel directions) is *not* part of the candidate argument.

Proposition B.1 (R29N.1: three outer-edge states)**.** *Under the far-end hypothesis of Proposition 7.1 with* TOP *abutting, the remaining outer-edge states are exactly: (i) LEFT gap owned by C, RIGHT and TOP abutting; (ii) RIGHT gap owned by C, LEFT and TOP abutting; (iii) LEFT, RIGHT and TOP all abutting.*

Proposition B.2 (R29N.2: nonempty $F \cap H$ forces C off BOTTOM)**.** *If $F \cap H \neq \varnothing$, then C avoids* BOTTOM *entirely.*

Proposition B.3 (R29N.4: triple abutment + nonempty $F \cap H$)**.** *Under triple abutment and $F \cap H \neq \varnothing$, the free corner W cannot contain the top mid. (Does not migrate to lateral-gap cases.)*

Full inequality chains, including the 266-coefficient adjacent-corner certificates retained from R13, are in `artifacts/closure/M2_R29_module_zh.md` and the long proof.

Appendix C. Closure patch M3: explicit branch tables

Class	Definition	Closing modules	Exhaustion note
G5	$F \ni O$; $k \in \{0,1,2\}$	R2–R8	Prop. 5.1
G1	unique centre, $k = 0$	R9–R10C	Prop. 5.2
centre-G2	$k = 1$, mid in C	R11–R12	Prop. 5.3
G4	$k = 2$, adjacent mids	R20–R23	Prop. 5.4
G3	$k = 2$, opposite mids	R24–R25	Prop. 5.5
residual G2	$k = 1$, mid only in F	near Prop. 6.1; far Prop. 7.1+M8–M11	Cor. 5.6

TABLE 1. Table M3.0 (top-level). Full near-end / far-end / joining tables: `artifacts/closure/M3_branch_table_zh.md`.

The Python checker `artifacts/closure/m3_branch_checker.py` verifies that the near-end Boolean grid (24 cells) is covered by live closing rows plus explicitly impossible cells. It is *not* a geometric proof. Running it on the shipped tree returns `PASS`.

APPENDIX D. CLOSURE PATCH M4: E/H/S CERTIFICATE TIERS

Computational exclusions in the long proof are tagged:

(E) exact certificates (Bernstein / Farkas / rational IBB / CAD);
(H) hand proofs with solver cross-check;
(S) dual-solver UNSAT (z3 and cvc5) without an independent exact certificate.

On the tracked unique-query inventory accompanying the peer-ready dump, roughly 60% are (E) or (H) and roughly 40% remain (S); exact counts drift as Item-3 upgrades land and are recorded in `artifacts/closure/M4_unsat_inventory.md`. **Class (S) must not be rewritten as machine-independent proofs.** External referees should sample (S) queries for modelling faithfulness even when both solvers report UNSAT.

REFERENCES

1. American Mathematical Society, *Guidelines for publishing computer-assisted papers*, https://www.ams.org/about-us/governance/policy-statements/guidelines-computer-assisted-proofs, 2023, Accessed 2026-10-20.
2. Fan Chung and Ron Graham, *Packing equal squares into a large square*, Journal of Combinatorial Theory, Series A **116** (2009), no. 6, 1167–1175, Asymptotic packing waste $W(x)$; cash challenges do not apply to $S(6) = 2$.
3. ———, *Efficient packings of unit squares in a large square*, Discrete & Computational Geometry **64** (2020), 690–699, Asymptotic regime; cited only to distinguish from $S(6) = 2$.
4. György Dósa, Zsolt Lángi, and Zsolt Tuza, *Covering a square by congruent squares*, 2026, States Conjecture 1.1 that $S(6) = 2$ (equivalently $\Pi(2) = 7$) without proof; proves several exact small values and boundary-covering analogues.
5. Erich Friedman, *Packing unit squares in squares: A survey and new results*, https://erich-friedman.github.io/papers/covering/covering.html, 2024, Online survey/compilation pages on square covering; accessed 2026-10-20.
6. Erich Friedman and Michael Paterson, *Covering squares with unit squares*, Geombinatorics **XV** (2006), no. 3, Discusses constructions and small-case covering phenomena related to $S(n)$.
7. Janusz Januszewski, *Covering a square of side $2 + \varepsilon$ with unit squares*, The American Mathematical Monthly **116** (2009), no. 2, 174–178.
8. Alexander Soifer, *Cover-up squared*, Geombinatorics (2005), Series of notes on covering a square of side $n+\varepsilon$ with unit squares; see also Open Problem Garden entry "Covering a square with unit squares".

9. ———, *Covering a square of side $n + \varepsilon$ with unit squares*, Journal of Combinatorial Theory, Series A **113** (2006), no. 2, 380–383.

School of Physics, Peking University
Email address: zhouningyu27@gmail.com
URL: https://github.com/ZhouNingyu327/S6-eq-2